Agentic Delegation and the Language Frontier

What the threshold model predicts—and what the design cannot identify

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August 29, 2026

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Quispe & Xu (2026) · arXiv:2605.25438

`github.com/gsaco/ai-03-quispe/tree/branch1`

The paper and the developer's problem

Question and economic mechanism

Question. Does an agent expand the set of programming languages in which a developer can ship code? This is a production frontier, not a claim that the developer learns each language.

Mechanism. Conversational AI augments existing language skill; agentic AI adds delegated execution under human specification and verification.

Choice problem for developer i, language k, month t

Pre-agent menu $\mathcal{M}_1 = \{S, C\}$; post-agent menu $\mathcal{M}_2 = \{S, C, D\}$:

$$m^*_{ikt} \in \arg\max_{m \in \mathcal{M}_g} V^m_{ikt}, \qquad Z^g_{ikt} = \mathbb{I}(\max_m V^m_{ikt} \geq 0).$$

Monthly breadth is $N^g_{it} = \sum_k Z^g_{ikt}$.

S**:** own execution.

C**:** suggestion gain $\gamma s - r_C$.

D**:** agent execution $\lambda a z(A)$, net of verification $\kappa(a, s)$, compute r_D, and residual risk.

Main result: a conditional activation band

Proposition 2 for an unfamiliar language

If conversational assistance has no value without a skill foothold, $\gamma \underline{s} - r_C \leq 0$, then $T^1 = T^S$. If delegation lowers the threshold, $B \equiv T^S - T^D > 0$, then

$$Z^2 - Z^1 = \mathbb{I}(T^D \leq \omega < T^S), \qquad \Pr(Z^2 - Z^1 = 1) = F(T^S) - F(T^D).$$

Conditions that do the work

Theory

- no conversational foothold in k;
- verification, compute, and residual-risk costs leave $B > 0$;
- opportunity mass lies inside $[T^D, T^S)$;
- the developer can specify and verify.

Empirical interpretation

- public, sustained adopters; "new" since Jan. 2024;
- not-yet adopters are valid counterfactuals;
- no time-varying project shock jointly moves adoption and languages.

Observed at adoption: +2.53 active languages and +1.19 newly used languages; these are event-time associations, not identified causal effects.

What I did: one modelling choice flips the conclusion

Quispe-Xu: production expands

Language skill s_{ik} is a **fixed state** over the choice horizon. Claude adds delegation to the menu and lowers the entry threshold:

$$\{S, C\} \longrightarrow \{S, C, D\}, \qquad T^D < T^S.$$

Result: more languages become feasible. This is explicitly a production frontier, not learning.

Aouad–Lykouris–Zhong: skills can erode

Skill, effort, and AI are substitutable inputs:

$$x_t = s_t + e_t + a_t, \qquad a_t \uparrow \Rightarrow e_t^* \downarrow.$$

They then make future skill depend on current effort. With myopic users and a positive sensitivity gap, lower effort shifts the steady-state skill distribution downward and productivity can fall.

The decisive choice is the skill law of motion: fixed skill gives static frontier expansion; effort-dependent skill turns substitution into deskilling.

Aouad, Lykouris & Zhong (2026), arXiv:2605.11350. With exogenous skill transitions, their productivity paradox disappears.

Hand and Lean check: where I did not believe the AI

Handwritten photo goes here

Add `hand/selection-derivation.jpg`

Copy the four-line derivation in `hand/README.md`.

Lean result 1 · threshold non-uniqueness

For every $T^D < T^1$, Lean constructs a generic gain $g > 0$ such that $T^1 - g = T^D$. The lower threshold and its activation band are therefore not uniquely agentic.

Lean result 2 · zero-effect selection

Lean also verifies the sign implication

$$\tau = 0,\ \beta > 0,\ P_A > P_C \quad \Longrightarrow \quad \widehat{\mathrm{ATT}}(0) > 0.$$

A project shock can create adoption and diversification together while all earlier differences remain flat.

Verdict: coherent mechanism, unidentified cause

Quispe–Xu can explain production expansion even while the Aouad–Lykouris–Zhong feedback predicts deskilling. The data measure the first outcome, not the missing skill stock, and voluntary adoption still confounds delegation with project selection.